Tohar Hermon | Computer Engineer

053-6201840 | ToharHermon959@gmail.com | LinkedIn Profile | GitHub Portfolio | Personal Website

PROFESSIONAL SUMMARY : Computer Engineering graduate (GPA 87) with a deep interest in low-level programming and hardware-software integration. Hands-on experience in compiler design, RISC-V architecture, and FPGA logic. Skilled in C/C++ and Python, with a strong grasp of operating systems, memory management, and microprocessors. Fast learner and Company Sergeant Major (Res.) with proven leadership and problem-solving skills in high-pressure environments.

TECHNICAL SKILLS

- **Low-Level & Hardware:** C, C++, RISC-V Assembly, Verilog (VHDL), FPGA, Microprocessors (8086, ESP32, ARM).
- **Embedded & IoT:** Raspberry Pi 5, Arduino, RTOS concepts, Device Drivers, OpenCV (Edge computing), RFID/Sensors.
- **Systems & OS:** Linux (Kernel/Shell), Memory Management, Multi-threading, Interrupt Handling, DSP.
- **Tools:** Git, Docker, Makefile, CMake, Logic Analyzers/Oscilloscopes (basic), MATLAB.
- **Programming:** Python, SQL, Java, OOP, Design Patterns.

TECHNICAL PROJECTS:

CPL to RISC-V Compiler | C, Lex, Bison, RISC-V Assembly

- Designed a full compilation pipeline translating a custom language (CPL) into RISC-V Assembly.
- Managed hardware constraints including register allocation, memory alignment, and instruction set mapping.
- Implemented Lexical/Syntax analysis and scope-aware symbol tables for low-level code generation.

Smart Door Lock System (IoT) | Python, OpenCV, Raspberry Pi 5

- Developed an Embedded security system featuring real-time face recognition and MFA via RFID/Sensors.
- Optimized performance for edge hardware, ensuring low-latency processing and efficient resource usage.
- Integrated hardware peripherals and implemented secure communication protocols between sensors and controllers.

7-Boom FPGA Game | Verilog, Altera/Cyclone

- Designed and implemented complex game logic on FPGA using Verilog.
- Applied timing analysis, clock synchronization, and debugging principles on physical hardware.

Cloud-Based CRM System | Next.js, AWS

- Developed a scalable serverless solution. Demonstrates broad architectural understanding and API integration skills.

WORK EXPERIENCE: Security Shift Supervisor | NVIDIA, Israel | 2021 – Present

- Managed multi-site security operations, leading teams of officers and ensuring 100% operational compliance.
- Resolved complex operational challenges through effective problem-solving and cross-departmental communication.

Company Sergeant Major (Res.) | IDF

- Managing logistics for 150 soldiers and maintaining operational readiness under intense pressure.
- Leading specialized sniper teams and coordinating high-stakes field operations.

EDUCATION : B.Sc. in Computer Engineering | Ruppin Academic Center | 2021 – 2025

- **GPA: 87**. |**Dean's List:** 2022/2023, 2023/2024, 2024/2025.
- **Relevant Coursework:** VHDL Design (100), Data Structures (100), Algorithms (95), Digital Signal Processing (91), Computer Architecture (90), Microprocessors (88).